PAPER_675: UQFF Analysis of GW170817: NS-NS Merger, GRB Delay, and Tidal Deformability

Author: Daniel Murphy
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Class: `UQFFComparedToGW170817` | CP4 #259
Source: `grok_share_fc21e30c24b4.txt`
Domain: Multi-Messenger Astrophysics

Abstract

GW170817, the first binary neutron star merger detected by LIGO/Virgo (August 17, 2017), provides the most stringent multi-messenger test of modified gravity. The UQFF predicts a modified GRB delay $\Delta t_{\text{UQFF}} = 1.7\,(1 + f_{\text{TRZ}}\rho_{\text{UA}}/\rho_{\text{SCm}})$ s and a reduced tidal deformability $\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}}(1 - f_{\text{TRZ}}\rho_{\text{SCm}}/\rho_{\text{UA}})$. Both effects are consistent with the observed 1.74 s EM/GW delay within UQFF parameter uncertainties.

Primary UQFF Equation

$$\Delta t_{\text{UQFF}} = 1.7\left(1 + f_{\text{TRZ}}\frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}\right)\text{ s}$$

Parameters

Parameter	Value	Description
m_1	$1.36\,M_\odot$	NS1 mass
m_2	$1.17\,M_\odot$	NS2 mass
d	$40\,\text{Mpc}$	Distance
f_{peak}	1500 Hz	Post-merger freq

UQFF Constants

Constant	Value	Description
ρ_{UA}	7.09×10^{-36} kg/m^3	Universal Aether density
ρ_{SCm}	7.09×10^{-37} kg/m^3	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	5×10^{-4} day^{-1}	UQFF calibration constant
$[\text{SSq}]$	0.57	Superstring quenching factor
μ_J	3.38×10^{23} J·m	Magnetic string coupling
γ	5×10^{-5} / 86400 s^{-1}	Aether oscillation decay

C++ Implementation

```cpp
#include "UQFFComparedToGW170817.h"

UQFFComparedToGW170817 obj;
auto result = obj.compute_primary(...); // primary equation
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```python
from CondensedPhysics2 import UQFFComparedToGW170817Calculator

calc = UQFFComparedToGW170817Calculator()
result = calc.compute()  # returns all UQFF quantities
print(result)
```

References

Abbott et al. 2017 (GW170817); Monitor of All-sky X-ray Image (MAXI); UQFF Session 173

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